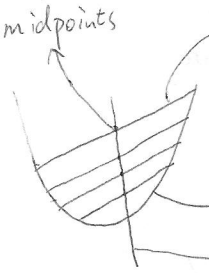

Quadratic form

a second-degree curve : $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$ → Linear form.

locus (轨迹) of the midpoints of the chords is a straight line

Parabola : $y=ax^2+bx+c$ (抛物线)

Circle : $(x-h)^2+(y-k)^2=r^2$

Ellipse : $\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$

Hyperbola : $\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$ 双曲线

$\underbrace{A(x,x)}_{\text{Quadratic form}}+\underbrace{L(x)}_{\text{Linear form}}+C=0$ 证明 locus 是 straight line : 线性.

Bilinear forms : linear numerical functions of two vector arguments.
$A(x,y)$

当 $x=y$ 时 Bilinear form 变成 Quadratic form.

Symmetric bilinear forms : $A(x,y)=A(y,x)$.

One of the basic problems of plane analytic geometry is to reduce the general equation of a second-degree curve to canonical form by transforming to a new coordinate system.

$Ax^2+\underbrace{2Bxy}_{\text{对称}}+Cy^2=D.$

$Q(x)=x^TAx$

为什么通常只考虑 A symmetric? 因为非对称部分会抵消.

$A=\frac{A+A^T}{2}+\frac{A-A^T}{2}$

$x^T(A-A^T)x=-x^T(A^T-A)x=-x^T(A-A^T)^Tx=-\underbrace{(x^T(A-A^T)x)}_{\text{常数 1-D 一维}}^T$

所以 $x^T(A-A^T)x=0$

$Q(x)=x^TAx=\frac{1}{2}x^T\underbrace{(A+A^T)}_{\text{symmetric}}x$

A is called the matrix of the quadratic form.

定义

- positive definite : $Q(x) > 0$ for all $x \neq 0$.
- negative definite : $Q(x) < 0$ for all $x \neq 0$
- indefinite : $Q(x)$ assumes both positive and negative values

positive $Q(x) \geq 0$ semi definite.

$Q(x) \leq 0$ negative semidefinite.

positive definite $\Leftrightarrow$ eigenvalues of A are all positive. } Theorem

$\boxed{\text{Principal Axes}}$ Theorem : there is an orthogonal change of variable, $x = Py$ that transforms the quadratic form $x^T A x$ into a quadratic form $y^T D y$ with no <u>cross-product</u> term.